

- Secure connectivity
- Smart factories
- Sustainable manufacturing
- Trustworthy AI with human expertise
- Workforce augmentation

Major applications of Industry 5.0 include 5G-enabled factories, AR-VR assisted operations, Industry 5.0 education labs, adaptive supply chains, aerospace production optimisation, climate-adaptive farming, demand forecasting, digital twins, digital workforce training, edge AI systems, energy optimisation, ethical AI deployment, explainable AI, food traceability, green factories, healthcare manufacturing, human well-being analytics, industrial cybersecurity, intelligent inventory management, last-mile optimisation, low-latency industrial networks, mass customisation, medical device personalisation, and mission-critical systems.

The market associated with Industry 5.0, which includes AI, augmented and virtual reality, digital twins, and smart robots, is predicted to grow over 34% between 2024 and 2032.

Industry 5.0 and AI integration

Industry 5.0 integrates AI with a preference for human inputs. Industry 4.0 involves complete automation for mass production, but since it does not focus on human feedback the outcomes are not reliable and are unacceptable in many cases. In Industry 5.0, AI is engaged to assist human experts rather than replace them. These experts take care of each phase of the Industry 5.0 process for a better and reliable final product.

Here's a brief description of two popular open source tools used for Industry 5.0.

OpenRemote

<https://openremote.io/>

OpenRemote is a powerful platform for smart automation and the Internet

Table 2: A comparison of Industry 4.0 and Industry 5.0

Aspect	Core focus	Industry 4.0	Industry 5.0
Vision	Human value	Smart automation	Human-centric
Role	Collaboration	Machine-driven	Human-guided
Technology	Intelligence	Cyber-physical systems	Cognitive systems
AI usage	Explainability	Decision automation	Decision support
Workforce	Empowerment	Skill replacement	Skill augmentation
Production	Personalisation	Mass production	Mass customisation
Systems	Control balance	Autonomous systems	Human-in-loop
Sustainability	Responsibility	Efficiency driven	Sustainability driven
Resilience	Continuity	Optimised flows	Adaptive systems
Ethics	Trustworthiness	Optional ethics	Embedded ethics
Data	Privacy	Centralised data	Sovereign data
Networks	Safety	Connected machines	Secure networks

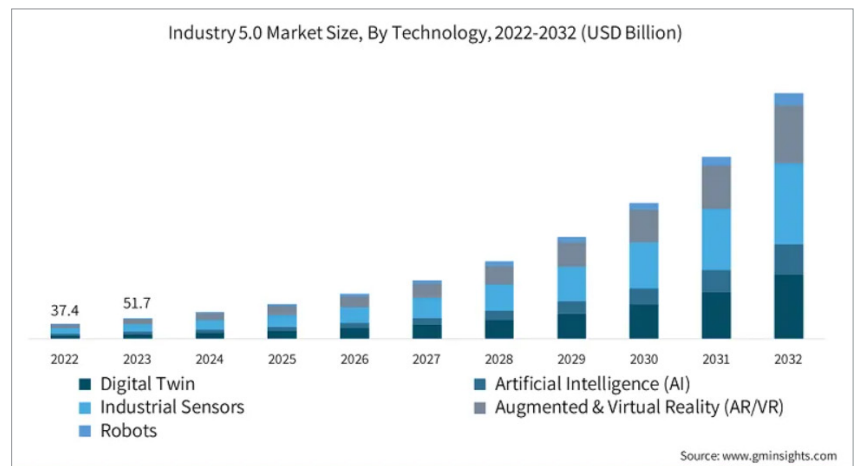


Figure 1: Market size of Industry 5.0 (Source: Global Market Insights)

Table 3: Popular tools for Industry 5.0 and AI integration

Tool	Domain	Purpose	URL
Webots	Robotics	Simulation	cyberbotics.com
ROS	Robotics	Control	ros.org
Gazebo	Robotics	Testing	gazebo.org
SUMO	Simulation	Mobility	eclipse.org/sumo
OpenRemote	IoT	Integration	openremote.io
Eclipse Ditto	Twins	Modelling	eclipse.org/ditto
TensorFlow	AI	Learning	tensorflow.org
PyTorch	AI	Training	pytorch.org
Hugging Face	LLMs	NLP	huggingface.co
KNIME	Analytics	Pipelines	knime.com
Apache Kafka	Streaming	Messaging	kafka.apache.org
Eclipse Mosquitto	IoT	Messaging	mosquitto.org